

①. $K > 0$, $\dot{x} = -K(x-21)$

$x(t) = \text{temp. tea}$

a) flow?

IVP: $\begin{cases} \dot{x} = -K(x-21) \\ x(0) = \eta \end{cases}$ linear non-hom. d.e.

$x(t)$ is st. c. $\Rightarrow \dot{x}(t) = 0$ $\Rightarrow -K(x-21) = 0 \Rightarrow x = 21 \Rightarrow x_p(t) = 21$.

Linear homog. eq. associated: $\dot{x} + Kx = 0$

$x_p(t) = 21$ part. sol.

$x + K = 0 \Rightarrow K = -K$

$\Rightarrow x_h(t) = c_1 \cdot e^{-Kt}$ gen sol.

$x(t) = x_h(t) + x_p(t) = c_1 \cdot e^{-Kt} + 21 \Rightarrow x(t) = (\eta - 21) \cdot e^{-Kt} + 21$

$x(0) = \eta = c_1 + 21 \Rightarrow c_1 = \eta - 21$

$\Rightarrow \varphi(t, \eta) = (\eta - 21) \cdot e^{-Kt} + 21$

b) $49^\circ\text{C} \xrightarrow{t=10} 37^\circ\text{C}$

? $\xrightarrow{t=20} 37^\circ\text{C}$

$= (49, 49) \xrightarrow{t=10} (49 - 21) \cdot e^{-K \cdot 10} + 21 = 37$

$\Rightarrow 28 \cdot e^{-10K} = 16$

$\Rightarrow 2e^{-10K} = 1$

$\Rightarrow e^{-10K} = \frac{1}{2}$

$\Rightarrow \frac{1}{e^{10K}} = \frac{1}{2}$

$\Rightarrow e^{10K} = 2 \Rightarrow K = \frac{\ln 2}{10}$

$\Rightarrow e^{10K} = e^{\ln 2} \Rightarrow 10K = \ln 2 \Rightarrow K = \frac{\ln 2}{10}$

② $\theta'' + g\theta = 0$, $\theta(0) = \frac{\pi}{2}$, $\theta'(0) = 0$.

Sol IVP?

after how much time pendulum $\rightarrow$ init. state?

$\theta(t)$ = angle at time t between rod, vertical.
(radians) (min)

char. eq: $\lambda^2 + g = 0$
 $\lambda = \pm 3i$

$$\theta(t) = c_1 \cos 3t + c_2 \sin 3t$$

$$\theta(0) = c_1 = \frac{\pi}{2}$$

$$\theta'(t) = -3c_1 \sin 3t + 3c_2 \cos 3t$$

$$\theta'(0) = 3c_2 = 0 \Rightarrow c_2 = 0$$

$$\theta(t) = \frac{\pi}{2} \cos 3t; \quad \theta'(t) = -3 \cdot \frac{\pi}{2} \sin 3t$$

Initial state: $\begin{cases} \theta(t) = \frac{\pi}{2} \\ \theta'(t) = 0 \end{cases} \Rightarrow \begin{cases} \frac{\pi}{2} \cos 3t = \frac{\pi}{2} \Rightarrow \cos 3t = 1 \\ -\frac{3\pi}{2} \sin 3t = 0 \Rightarrow \sin 3t = 0 \end{cases}$

$$\Rightarrow 3t = 2k\pi \Rightarrow t = \frac{2k\pi}{3}$$

$$\theta(t) = \frac{\pi}{2} \cos 3t$$

pendulum returns to initial state after $\frac{2\pi}{3}$ min.

③ $g: I \rightarrow \mathbb{R}$, g' map s.t. $g'(x) \neq 0$, $\forall x \in I$.

Assume $\exists \kappa \in I$ s.t. $g(\kappa) = 0$

? for $\eta \in I$ sufficiently close to $\kappa \Rightarrow$ the unique sol. $(x_k)_{k \geq 0}$ of IVP

$$x_{k+1} = x_k - \frac{g(x_k)}{g'(x_k)}, \quad x_0 = \eta \text{ satisfies } \lim_{k \rightarrow \infty} x_k = \kappa.$$

$$\text{Let } F(x) = x - \frac{g(x)}{g'(x)} = x_{k+1}$$

$$g(\kappa) = 0 \Rightarrow F(\kappa) = \kappa - 0 = \kappa. \Rightarrow \kappa \text{ fixed point}$$

Mean Value Theorem: $\underbrace{g(x) - g(\kappa)}_0 = g'(c)(x - \kappa), \quad c \in (x, \kappa)$

$$\Rightarrow g(x) = g'(c)(x - \kappa)$$

$$\begin{aligned} F(x) - F(\kappa) &= F(x) - \kappa = x - \frac{g(x)}{g'(x)} - \kappa = x - \kappa - \frac{g'(c)(x - \kappa)}{g'(x)} = \\ &= (x - \kappa) \left(1 - \frac{g'(c)}{g'(x)} \right) \end{aligned}$$

$$g' \text{ continuous, } g'(\kappa) \neq 0 \Rightarrow \frac{g'(c)}{g'(x)} \rightarrow 1 \text{ as } x \rightarrow \kappa. \Rightarrow$$

$$\frac{|F(x) - \eta|}{|x - \eta|} \rightarrow 0$$

$$|F(x) - \eta| \leq g|x - \eta|, \quad 0 < g < 1.$$

$$x_0 = \eta \Rightarrow |x_{k+1} - \eta| \leq g|x_k - \eta|$$

$$\Rightarrow |x_k - \eta| \leq g^k |\eta - \eta| \quad \underline{\underline{g^k \rightarrow 0}}$$

$$\Rightarrow \lim_{k \rightarrow \infty} x_k = \eta$$

$$\textcircled{4} \text{ IVP: } \begin{cases} y' = 1 + xy^2 \\ y(0) = 0 \quad (\Leftrightarrow y_0 = 0) \\ \quad \quad \quad (k=0) \end{cases}$$

Euler on $[0, 1]$, $h = 0.02$

- init. values?
- nr. steps to find approx of $\varphi(0.5)$? and $\varphi(1)$
approx. of $\varphi(0.04)$?

$$x_k = k \cdot h = 0.02 \cdot k$$

$$y_{k+1} = y_k + h \cdot y_k' = y_k + 0.02 \cdot (1 + \tilde{x}_k \cdot y_k^2) = y_k + 0.02 \cdot (1 + 0.02 \cdot k \cdot y_k^2)$$

$$\text{let } y_k = f(x, y) = 1 + x y^2$$

$$x_0 = 0.02 \cdot 0 = 0 \quad (k=0)$$

$$\Rightarrow \text{initial values: } x_0 = 0, y_0 = 0.$$

$$y_0 = y(0) = 0. \quad (k=0)$$

$$\begin{aligned} \bullet \varphi(0.5) &\Rightarrow x_k = 0.5 = 0.02 \cdot k \Rightarrow k = \frac{0.5}{0.02} = \frac{\frac{5}{10}}{\frac{2}{100}} = \frac{50}{2} = 25 \\ &\Rightarrow \underline{\varphi(0.5) \text{ approximated by } y_{25}} \end{aligned}$$

$$\bullet \varphi(1) \Rightarrow x_k = 1 = 0.02 \cdot k \Rightarrow k = \frac{1}{0.02} = \frac{1}{\frac{2}{100}} = 50.$$

$$\Rightarrow \underline{\varphi(1) \text{ approximated by } y_{50}}$$

$$\bullet \varphi(0.04) \Rightarrow 0.04 = 2 \cdot 0.02 \Rightarrow k = 2 \Rightarrow y_2.$$

$$x_0 = 0 \quad y_0 = 0 \quad \Rightarrow y_1 = y_0 + 0.02 \cdot (1 + x_0 y_0^2) = 0 + 0.02 \cdot (1 + 0) = 0.02$$

$$x_1 = 0.02 \quad y_1 = 0.02 \quad \Rightarrow y_2 = y_1 + 0.02 \cdot (1 + x_1 y_1^2) = 0.02 + 0.02 \cdot (1 + 0.02 \cdot (0.02)^2)$$

$$y_2 = 0.040004 \quad 0.040004 \dots$$